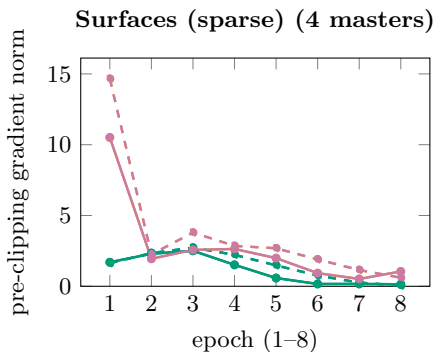
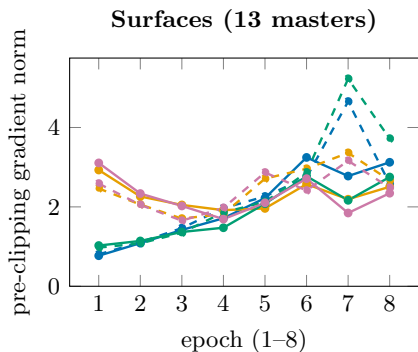
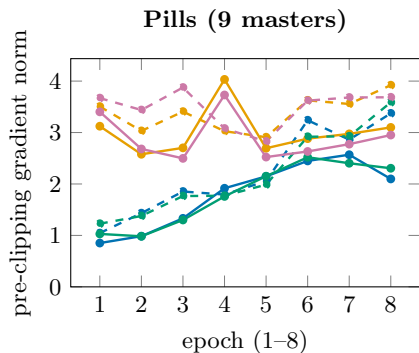
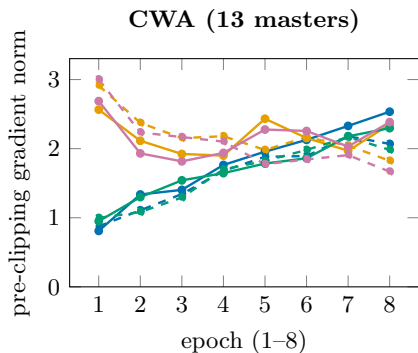


P05-S02 pre-clipping gradient norm over epochs 1-8

Eight-epoch training-only numerical check; no unseen-data comparison.



Solid line: seed1; dashed line: seed2. D0-M ordinary classification; D1 chemical similarity; D2 matched-sample consistency; D3 both. Sparse panel: cross-instrument measurements of the same sample are unavailable; D2/D0-M and D3/D1 overlap by construction; overlap is not evidence of learned invariance.

— D0-M ordinary classification — D1 chemical similarity
— D2 matched-sample consistency — D3 both
— seed1 (solid) - - - seed2 (dashed)